

Doc link:-

https://docs.google.com/document/d/1vO6f_jiVCr2Ql-Bm6pw8V_n6C-idHCFt9Jm4Gfv3uHM/edit?usp=sharing

Chapter link:- <https://jax-ml.github.io/scaling-book/gpus/>

Summary of GPU specs

Here is a summary of GPU specs for recent models. The number of SMs, clock speed, and FLOPs differ somewhat between variants of a given GPU. Here are memory capacity numbers:

GPU	Generation	Clock Speed	SMs/chip	SMEM capacity/SM	L2 capacity/chip	HBM capacity/chip
V100	Volta	1.25GHz/1.38GHz	80	96kB	6MB	32GB
A100	Ampere	1.10GHz/1.41GHz	108	192kB	40MB	80GB
H100	Hopper	1.59GHz/1.98GHz	132	256kB	50MB	80GB
H200	Hopper	1.59GHz/1.98GHz	132	256kB	50MB	141GB
B200	Blackwell	?	148	256kB	126MB	192GB

All generations have 256kB of register memory per SM. Blackwell adds 256kB of TMEM per SM as well. Here are the FLOPs and bandwidth numbers for each chip:

GPU	Generation	HBM BW/chip	FLOPs/s/chip (bf16/fp16)	FLOPs/s/chip (fp8/int8)	FLOPs/s/chip (fp4)
V100	Volta	9.0e11	—	—	—
A100	Ampere	2.0e12	3.1e14	6.2e14	—
H100	Hopper	3.4e12	9.9e14	2.0e15	—

H200	Hopper	4.8e12	9.9e14	2.0e15	—
B200	Blackwell	8.0e12	2.3e15	4.5e15	9.0e15

Question 1 [CUDA cores]: How many fp32 CUDA cores (ALUs) does an H100 have? B200? How does this compare to the number of independent ALUs in a TPU v5p?

My Answer:-

H100=132*4*32=16896

B200= 148*4*32=18944

TPU=cores*lanes*sublanes*[no.of](#) ALU =2*8*128*4=8192

Question 2 [Vector FLOPs calculation]: A single H100 has 132 SMs and runs at a clock speed of 1.59GHz (up to 1.98GHz boost). Assume it can do one vector op per cycle per ALU. How many vector fp32 FLOPs can be done per second? With boost? How does this compare to matmul FLOPs?

My Answer:-

Normal = ALU count * freq=1.69e4*1.59e9=27 TFLOPs

boost=27/1.59*1.98=33.46 TFLOPs

MatMul FLOPs=990 TFLOPs

Question 3 [GPU matmul intensity]: What is the peak fp16 matmul intensity on an H100? A B200? What about fp8? *By intensity we mean the ratio of matmul FLOPs/s to memory bandwidth.*

My Answer:-

H100=9.9e14/3.4e12=291

B200=287.5

for fp8 it will get double

Question 4 [Matmul runtime]: Using the answer to Question 3, how long would you expect an `fp16[64, 4096] * fp16[4096, 8192]` matmul to take on a single B200? How about `fp16[512, 4096] * fp16[4096, 8192]`?

My Answer:-

BS<288 so HBM bound

$(2BD+2DF+2BF)/HBMBW=6.87e7/8e12=8.6$ micro sec

for other its compute bound so 15 micro second

Question 5 [L1 cache capacity]: What is the total L1/SMEM capacity for an H100? What about register memory? How does this compare to TPU VMEM capacity?

My Answer:-

$L1=132*256KB=33MB$

register=132*256KB=33 MB

Question 6 [Calculating B200 clock frequency]: NVIDIA reports [here](#) that a B200 can perform 80TFLOPs/s of vector fp32 compute. Given that each CUDA core can perform 2 FLOPs/cycle in a FMA (fused multiply add) op, estimate the peak clock cycle.

My Answer:-

ops per sec=ALU*2*freq

so total fres=80e12/3.8e4=2.1GHz

Question 7 [Estimating H100 add runtime]: Using the figures above, calculate how long it ought to take to add two `fp32[N]` vectors together on a single H100. Calculate both T_{math} and T_{comm} . What is the arithmetic intensity of this operation? If you can get access, try running this operation in PyTorch or JAX as well for `N = 1024` and `N=1024 * 1024 * 1024`. How does this compare?

My Answer:-

$T_{math}=N^2/80e12$

$T_{comm}=N^4/3/3.4e12$

so comm bound

3.53 ms for n=1024

Networking

NVLink Gen	NVSwitch Gen	GPU Generation	NVLink Bandwidth (GB/s, full-duplex)	NVLink Ports / GPU	Node GPU to GPU bandwidth (GB/s full-duplex)	Node size (NVLink domain)	NVSwitches per node
3.0	2.0	Ampere	25	12	300	8	6
4.0	3.0	Hopper	25	18	450	8	4
5.0	4.0	Blackwell	50	18	900	8/72	2/18

Quiz 2: GPU nodes

Here are some more Q/A problems on networking. I find these particularly useful to do out, since they make you work through the actual communication patterns.

Question 1 [Total bandwidth for H100 node]: How much total bandwidth do we have per node in an 8xH100 node with 4 switches? *Hint*: consider both the NVLink and NVSwitch bandwidth.

My Answer:-

speed=min(switch speed,gpu link speed)

=min(numswitches*numports*speedperport,numGPUs*numlinks per gpu* link speed)

=min(4*64*25e9,8*18*25e9)

=3.6TBps

Question 2 [Bisection bandwidth]: Bisection bandwidth is defined as the smallest bandwidth available between any even partition of a network. In other words, if we split a network into two equal halves, how much bandwidth crosses between the two halves? Can you calculate the bisection bandwidth of an 8x H100 node? *Hint*: bisection bandwidth typically includes flow in both directions.

My Answer:-

one GPU can send at 450GBps and 4 and 4 gpu in 2 partetion and as this is full duplex speed we can have total flow of $450 \times 8 = 3.6$ TBps

Question 3 [AllGather cost]: Given an array of B bytes, how long would a (throughput-bound) AllGather take on an 8xH100 node? Do the math for bf16[Dx, F] where $D=4096$, $F=65,536$. It's worth reading the TPU collectives [section](#) before answering this. Think this through here but we'll talk much more about collectives next.

My Answer:-

per node speed= $|v|/N$

coms speed per link= W_{link}

no. of sends= $n-1$

no. of receives= $n-1$ total full duplex

so total time $= (n-1) * |v| / n / w_{link} = 1.04ms$

Pop Quiz 1 [AllGather time]: Using an 8xH100 node with 450 GB/s full-duplex bandwidth, how long does AllGather(bf16[Bx, F]) take? Let $B=1024$ $F=16,384$

My Answer:-

$(n-1) * BF^2 / (n * \text{Full duplex BW}) = 75 \text{ micro sec}$

Pop Quiz 2 [AllToAll time]: Using an 8xH100 node with 450 GB/s unidirectional bandwidth, how long does AllToAll $_{x \rightarrow N}$ (bf16[Bx, N]) take? What if we know only 4 of 8 entries will be non-zero?

My answer it is around $\min(k/N, 1) * (n-1)BN^2 / (N * N * BW)$

Question 1 [SU AllGather]: Consider only a single SU with M nodes and N GPUs per node. Precisely how many bytes are ingressed and egressed by the node level switch during an AllGather? What about the top-level switch?

My answer:-

for node level ingress takes in $N * B / MN$ and $N * (B - B / MN)$ egress to gpu

then for leaf it egress B / M and ingress B so

limiting is node egress of NB so it is NB / W_{node}

for sapine the limit time is $M * B(M-1) / M$

Question 2 [Single-node SHARP AR]: Consider a single node with N GPUs per node. Precisely how many bytes are ingressed and egressed by the switch during an AllReduce using SHARP (in-network reductions)?

My answer:-

for all reduce the node sends $(n-1)B/n$ at first then receives B/n then sends B/n and again receives $(n-1)B/n$ so total egress and ingress $=B$ bytes

Question 3 [Cross-node SHARP AR]: Consider an array $bf16[Dx, Fy]$ sharded over a single node of N GPUs. How long does $AllReduce(bf16[D, Fy] \{ Ux \})$ take? You can assume we do in-network reductions. Explain how this differs if we have more than a single node?

My answer:-

because everything is on one node then total B/Y and receives $B/Y=2DF/YW_{gpu}$

Question 4 [Spine level AR cost]: Consider the same setting as above, but with $Y=256$ (so the AR happens at the spine level). How long does the AllReduce take? Again, feel free to assume in-network reductions.

My answer:-

at node level

each gpu sends B/XY to node switch and then receives B/Y so for gpu $t_{comm}=B/W_{gpu}Y$

for node level

it egresses at $8B/XY$ and ingresses $8B/Y$ so the time is $8B/YW_{switches}$

Question 5 [2-way AllGather cost]: Calculate the precise cost of an AllGather of B bytes over exactly 2 nodes. *Make sure to calculate the precise cost and not the approximation, and consider both the intra-node and cross-node cost.*

My answer:-

node to node is $B/2W_{node}$

and because of ring circulation of data in nodes it takes $7/8B/2$ time for both halves so the time taken is $7/8*(B/W_{gpu})$

Quiz 5: LLM rooflines

Question 1 [B200 rooflines]: A B200 DGX SuperPod (not GB200 NVL72) has 2x the bandwidth within a node (900GB/s egress) but the same amount of bandwidth in the scale-out network (400GB/s) ([source](#)). The total FLOPs are reported above. How does this change the model and data parallel rooflines?

My Answer:-

FLOPs= 2250 from 900

Bandwidth is doubled

for data parallelism so the data parallelism we need 2.5/2~1.25 time more number of tokens this is the intra node condition for inter node it will hurt our token count a lot by 2.5 times

For TP

for inside one node again a 1.25 times increase in batch size requires

for all inter node communication it will hurt our batch size by 2.5 times

Question 2 [How to shard LLaMA-3 70B]: Consider LLaMA-3 70B, training in bfloat16 with fp32 optimizer state with Adam.

1. At a minimum, how many H100s would we need simply to store the weights and optimizer?
2. Say we want to train on 4096 H100 GPUs for 15T tokens. Say we achieved 45% MFU (Model FLOPs Utilization). How long would it take to train?
3. LLaMA-3 70B has $F = 28,672$ and was trained with a batch size of about 4M tokens. What is the most model parallelism we could do without being comms-bound? With this plus pure DP, could we train LLaMA-3 while staying compute-bound on 4k chips? What about ZeRO-3? What about with 8-way pipelining? *Note: consider both the communication cost and GPU memory usage.*

My Answer:-

1

around 700GB including optimiser

per H100 memory=80 sor around 9 GPU just to store model

2

amount of flops roughly= $6 \times \text{number of tokens} \times \text{number of parameters} = 6.3e24$ FLOP

per GPU=990e12 FLOP/s

so time is around $6.3e24 / (0.45 \times 4096 \times 990e12) = 3.45e6 = 40$ days

3

$Y < F \times C / W$ W is 450 GBps inside node and 400 outside and C=990 TFLOPs so around 14 so like 8 way model parallelism

so like 512 for DP

but as we said 8 GPU won't fit model in 1 so like 16 GPU TP and 256 GPU DP but with **ZeRO-3** it **will work**

technically the BW in 8 ways pipelineing increases by

so out BW increases by 8 times so now we can do Batch size per GPU like in 990

$TFLOPS / (8 \times 400GBps) = 312$ batch size